

Lessons from Real-Time and Archival Acoustic Detections of North Atlantic Right Whales in the Southeast US across Three Types of Platforms

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The critically endangered North Atlantic right whale (NARW) migrates seasonally along the US East coast to their winter calving grounds in the Southeast US. In the shallow waters off the coasts of South Carolina, Georgia, and Florida, NARWs nurse, rest, and socialize near the surface, exposing whales and calves to a high risk of ship strike. In NARW's northern foraging grounds, real-time passive acoustic monitoring (PAM) using gliders and buoys is being used to inform monitoring efforts and communicate detections with vessel captains and managers to reduce risk of vessel strikes. In the Southeast US, glider-based PAM may also be a useful monitoring tool for providing 24/7 coverage. However, the unique shallow environment creates uncertainty in the detectability of NARWs in the region. To better inform the deployment of real-time instruments in this region, we analyzed acoustic data from three different platform types. Three bottom-mounted, archival hydrophones were deployed at increasing depths (15.9m, 19.5m, 22m) and distances off the coast of Savannah, Georgia, from January-May 2023-2026. Real-time PAM buoys were deployed off the coasts of Norfolk, Virginia and Savannah, Georgia at the beginning of the 2023 calving season. A real-time PAM-equipped Slocum Glider was deployed over 6 approximately 1-month long missions in January-March from 2023-2026. The Norfolk buoy triggered 17 acoustically-detected Slow Zones over this period, and 2 acoustic right whale detections on the Savannah buoy in January 2026 triggered the first ever acoustic Slow Zone in the South Atlantic Bight. The NOAA-issued Slow Zones provide critical real-time information to mariners about right whale presence and vessel strike risk, requesting reductions in vessel speeds to <10 knots. These detections and associated Slow Zone notices demonstrate potential for an acoustic real-time monitoring program that would complement existing aerial monitoring efforts in the Southeast US.